

Abdulla Sajad

Software Engineer - Backend Systems and AI Infrastructure

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Summary

Software Engineer building backend systems first (FastAPI, PostgreSQL, Redis), then layering retrieval and AI on top. Shipped webhook delivery infrastructure, a document-retrieval backend, an open-source PyPI storage package, and a multi-service hiring platform.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL

Backend: FastAPI, REST APIs, Async Programming, SQLAlchemy, JWT, RBAC

Databases: PostgreSQL, Redis, Database Design, Indexing, Transactions

AI/Retrieval: RAG, Vector Search (FAISS), Embeddings, OpenAI APIs, Evaluation

DevOps: Docker, GitHub Actions, CI/CD, Linux, Monitoring & Observability (Prometheus)

Projects

HookFlow - Webhook/Event Delivery Infrastructure | Python, FastAPI, PostgreSQL, Redis, Docker, Next.js | GitHub

- Built persist-first ingest: Event + Delivery rows committed in one transaction before any queue touch, so a lost Redis message costs latency, never an event.
- Enforced at-least-once safety with UNIQUE(endpoint_id, idempotency_key) plus atomic worker claims (FOR UPDATE SKIP LOCKED, 60s leases) - verified zero double delivery under concurrent workers; 73 tests.
- Added HMAC-signed delivery, exponential backoff, DLQ with replay, tenant isolation via per-tenant API keys, Alembic migrations, Prometheus metrics, and a Next.js operator dashboard.

Lexora AI - Document Processing and Retrieval Backend | Python, FastAPI, LangChain, OpenAI, FAISS, Redis, Celery, Prometheus | GitHub

- Engineered JWT auth with Redis-backed token revocation and refresh rotation; Redis retrieval caching; Celery async ingestion pipeline; 35-test suite.
- Built per-user FAISS index isolation with retrieval + citations, SSE streaming, and conversation history (pgvector migration planned and documented).

neuDB - Embedded Python Storage Engine | Python, FastAPI, PyPI | GitHub, PyPI

- Built and published a zero-dependency, file-based storage engine to PyPI with crash-atomic writes (temp + fsync + replace), thread-serialized access, and cosine-similarity semantic search with text fallback; 32 tests with measured O(N) benchmarks.
- Added session/memory schemas and a FastAPI HTTP layer used for agent-memory experiments.

EsenceLab - AI Hiring Platform | Next.js, TypeScript, Express, FastAPI, Supabase | GitHub, Live

- Built a 3-service system (Next.js, Express, FastAPI) with JWT auth and RBAC across student, recruiter, and admin roles, resume parsing and skill-matching, rate limiting and audit logging.
- Led a 3-engineer team; won 2nd Prize at Innovision Project Competition (30+ teams).

Experience

Independent Software Engineer - Remote (2023 - Present)

- Built and deployed an order-management and real-time tracking system (Next.js, Express, PostgreSQL, Supabase) for a college institution as sole engineer, replacing manual spreadsheet workflows.
- Built and deployed an e-commerce storefront (Next.js, Express, Supabase) for a local retail brand with catalog management and secure checkout.

Achievements

- 2nd Prize, Innovision Project Competition - led 3-engineer team building EsenceLab (30+ teams).
- Published neuDB, an open-source Python storage engine, on PyPI.

Education

Sree Narayana Guru College of Engineering and Technology - B.Tech, Computer Science and Engineering, CGPA 7.1/10 (2022 - 2026), Kerala, India

Vadi Husna Public School - Higher Secondary, Computer Science (2020 - 2022), Kerala, India